OIL PRICE FORECASTING

A Comparative Study of ARIMA and LSTM Models with 24-Month Forecasting Horizon

Advanced Research Topics in Data Science

Module Assignment Report

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Github Link : https://github.com/Sravani8787/Advanced_Research

1. Introduction

Crude oil prices are major indicators of the economy of the world that affect transportation, manufacturing, inflation and policy making. Nevertheless, oil prices are very volatile and non-stationary and are also sensitive to external shocks, including geopolitical events and OPEC decisions, which makes it difficult to predict them.

In this report, the comparison will be made between two forecasting methods: ARIMA in statistics and the neural network of LSTM in deep learning. Both models are projected and assessed using day-to-day oil price information (500 observations) during the period between September 2024 and February 2026. The paper evaluates their performance and identifies the approach, which is more adequate in this forecasting.

2. Literature Review

2.1 ARIMA in oil prices forecasting.
Another univariate time-series forecasting technique that is popularly used is the ARIMA model, proposed by George E. P. (*Box, G.E.P. and Jenkins, G.M. (1976) Time Series Analysis Forecasting and Control. 2nd Edition, Holden-Day, S. Francisco. - References - Scientific Research Publishing*, no date) It is a combination of autoregressive terms, moving average components, and differsencing to facilitate the stationarity. It has been demonstrated that ARIMA is better at short-to-middle term crude oil forecasts when price behaviour is fairly linear (Mohammadi and Su, 2010) Nevertheless, its primary weakness is that it assumes that the market is linear, which can decrease the accuracy in the times of market instability or structural shifts (Baumeister and Kilian, 2015)

2.2 Predicting Time Series -LSTM Networks.
Long Short-term memory (LSTM) networks Long-term memory (LSTM) networks are a category of recurrent neural net that attempts to learn long-term dependencies in sequential data. LSTMs are able to learn non-linear and complicated trends in time-series data, through gating techniques that filter flow of information. It has been demonstrated that LSTM models tend to be more useful than traditional statistical models in predicting the oil prices due to their ability to capture the non-linear market behaviour and regime shifts (Cen and Wang, 2019).

2.3 Comparative Studies
Some of the works provide a comparison between the statistical and deep learning models of oil price prediction. It has been found that ARIMA is effective in the very short term forecast, whereas the LSTM models have been known to perform better at longer horizons because they are capable of capturing complex patterns (Aamir, Shabri and Ishaq, 2018). Nevertheless, LSTM models need more significant datasets and computational power (Drachal, 2021) This is the reason why the current study is inspired by the need to compare the two methods with the latest oil price data of the year 2024-2026.

3. Data Description and Exploratory Analysis

The data is a 500 observations of daily oil prices between 21 st September 2024 and 2 February 2026, which is obtained in the additional material given as a module. The minimum price is 16.48

and the maximum price is 158.78 with the mean of 75.87 and the standard deviation of 42.91 showing that there is a high variation in prices over the period of observation.

Statistic	Value	Interpretation
Observations	500	Daily frequency
Mean Price	$75.87	Central tendency
Std. Deviation	$42.91	High volatility
Min / Max	$16.48 / $158.78	Wide range ($142.30)
Date Range	Sep 2024 – Feb 2026	~17 months

Table 1: Summary statistics of the oil price dataset.

The first exploratory visualisation is summarised in figure 1. The time series plot (top-left) shows a strong trend, which is upwards with no apparent daily level seasonality. The price distribution (top-right) is quite skewed towards the right, as it is quite expected by the rising trend in the prices during the period of observation. The boxplot by year (bottom-left) ensures that the price trend is indeed increasing, and the rolling 30-day statistics (bottom-right) indicates that neither the mean nor the standard deviation is constant showing a visual representation of non-stationarity at the early stage.

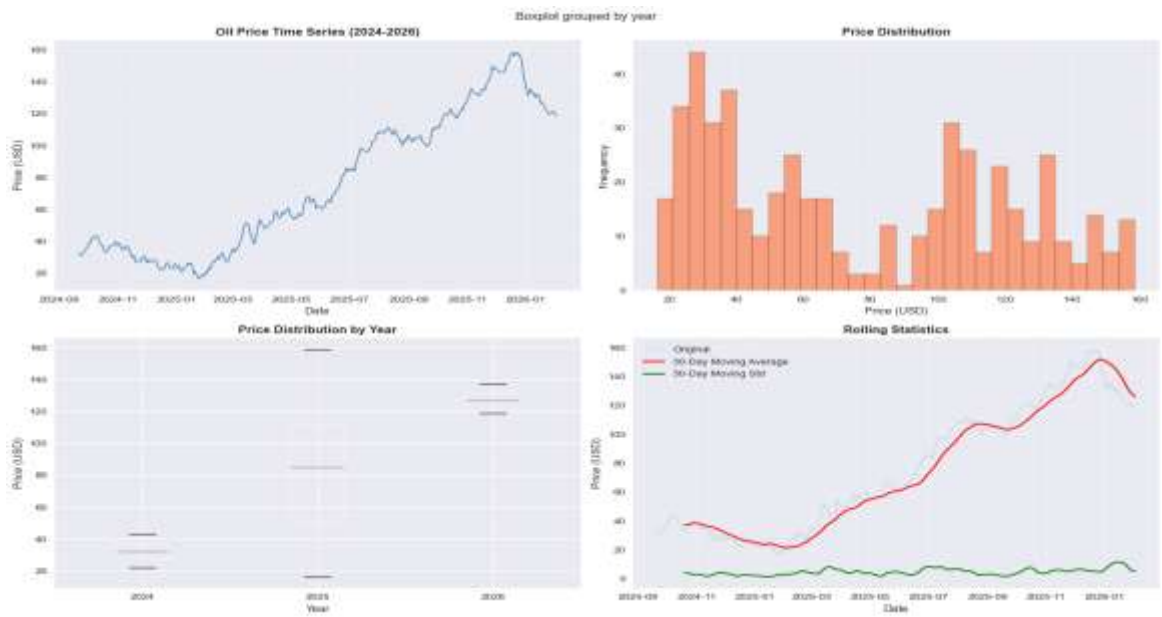

Figure 1: Exploratory data analysis — time series, distribution, yearly boxplots, and rolling statistics.

4. Stationarity Analysis and Data Preparation

ARIMA modelling has the prerequisite of stationarity. Constant mean, variance, and autocovariance of a time series make the series stationary. The stationarity was determined using two formal statistics tests.

The Augmented Dickey-Fuller (ADF) test had a test statistic of -0.546 and the p-value of 0.883, which was not less than the significance level of 0.05. At the same time, KPSS test gave a value of

3.706 with a p-value of less than 0.01 and it rejected the null that stationarity. The two tests have been unanimous in the fact that the original series is non-stationary and that they have to be differenced before ARIMA modelling.

Test	Statistic	p-value	Conclusion
ADF Test	−0.546	0.883	Non-stationary
KPSS Test	3.706	< 0.01	Non-stationary

Table 2: Stationarity test results on the original price series.

The autocorrelation function (ACF) of the original series shows a slow, gradual decay characteristic of non-stationary data, while the partial ACF (PACF) shows a sharp cutoff after lag 1, further supporting the presence of a unit root.

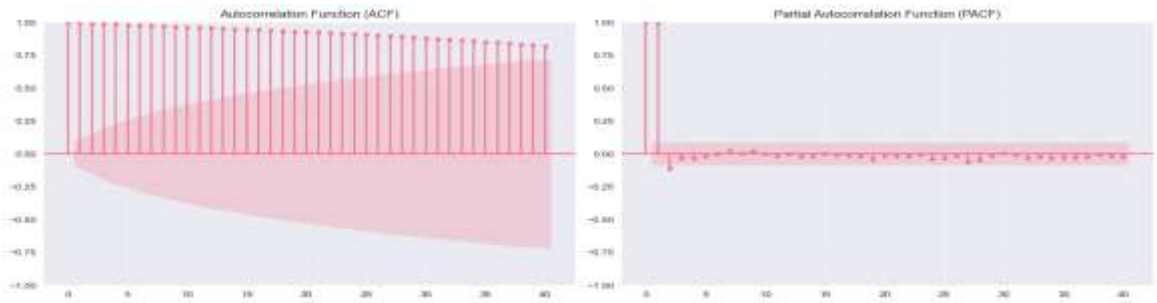

Figure 2: ACF and PACF of the original (non-stationary) oil price series.

First differencing was used, and the series obtained was re-tested. The first-differenced series ADF test yielded a very significant value (p-value much less than 0.05) which showed that first-order differencing is enough to obtain stationarity. Figure 3 compares the original series, first-differenced series and the second-differenced series.

Figure 3: Original, first-differenced, and second-differenced oil price series.

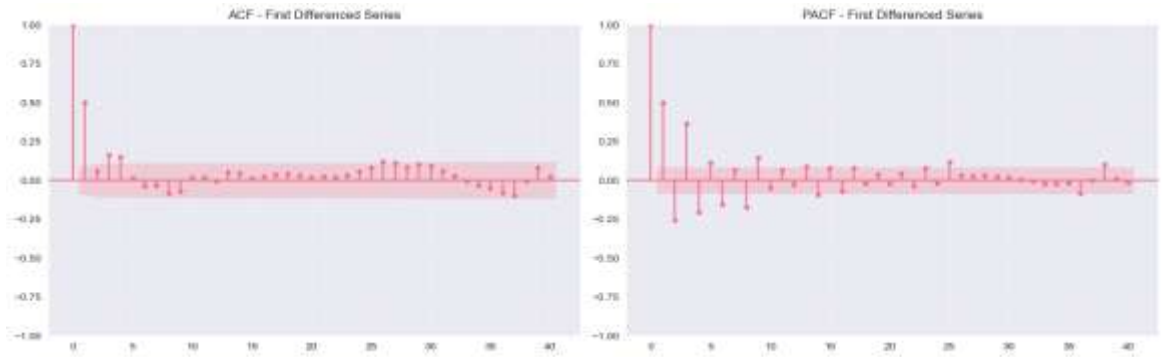

Figure 4: ACF and PACF of the first-differenced series — used to inform initial ARIMA parameter selection.

5. ARIMA Model Development and Evaluation

5.1 Parameter Optimisation

A grid search was used to exhaustively search all the combinations of parameters with p [0, 8], d [0, 2], q [0, 8], and a total of 729 candidate models were identified. Each model was fitted with the help of Maximum Likelihood Estimation (MLE) and the selection criterion was Akaike Information Criterion (AIC) as it takes into consideration the goodness of fit in a model and its complexity. The best model that was found was ARIMA(2, 2, 7) with an AIC of 1429.22 and BIC of 1471.32.

The fact that d = 2 (second-order differencing) has been chosen indicates that the series may have a stochastic trend which must be removed using two rounds of differencing. The autoregressive structure of short term is captured by the AR(2) component but the autoregressive structure at long term is captured by the MA(7) component which captures the effect of the past forecast errors to a maximum of seven lags. The comparatively complicated MA order can indicate the random noise-induce movement of the commodity prices.

5.2 Model Diagnostics

The systematic patterns in the data should be sufficiently well captured by the model, which is only possible through thorough residual analysis. The diagnostic suite is shown in four panels in figure 5. The remaining time series (top-left) oscillates around zero in a random way with neither visible trends nor patterns. The histogram (top-right) demonstrates a more or less Gaussian distribution. The Q-Q plot (bottom-left) proves the existence of almost perfect normality, as the residues are very close to the theoretical line. The ACF of the residuals (bottom-right) does not exhibit any significant level of autocorrelation with the lag, which proves that the residual values are essentially white noise.

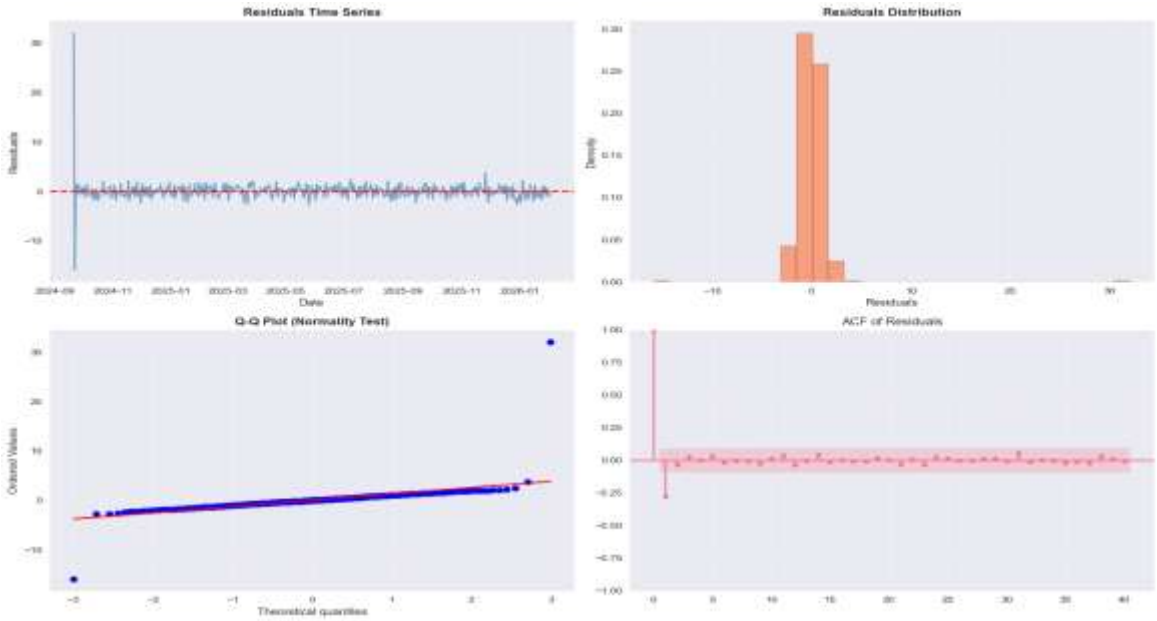

Figure 5: ARIMA(2,2,7) residual diagnostics — time series, histogram, Q-Q plot, and ACF.

The Ljung-Box test gave a Q-statistic of 0.00 at the lag number of 1 where the p-value of 0.96 was obtained, verifying the non-existence of serial correlation in the residuals. Shapiro-Wilk test of normality showed p-value of 0.97, which represents that the residuals are normally distributed. All these diagnostics indicate that ARIMA(2,2,7) model fits the observed data very well.

5.3 In-Sample Performance and Forecast

The model had an in-sample RMSE of 1.8850, MAE of 0.8875 and MAPE of 1.83 per cent and this shows that the model fits the historical data very well. The visual confirmation of the tight correspondence in actual and fitted values is made by Figure 6.

Figure 6: ARIMA model fit — actual versus predicted oil prices over the training period.

A 24 month (730days) prediction was then done using 95 percent confidence levels. ARIMA model predicts that the trend will increase, and mean forecast will be in the area of $245.63 in February 2028. The confidence intervals however blur significantly as the forecast horizon is reached, as a result of the compounding uncertainty of extrapolating to the long term. This is one of the well known limitations of ARIMA models because the variance of multi-step-ahead forecasts increases unlimitedly.

Figure 7: ARIMA 24-month oil price forecast with 95% confidence intervals.

6. LSTM Model Development and Evaluation

6.1 Architecture and Data preparation.
The LSTM net was built in both TensorFlow and Keras in a stacked network with 2 layers of LSTM (50 units each) with 20 percent Dropout regularisation layers. The transformation of features is done through a dense layer of 25 units followed by the single-unit eventual output layer that predicts the price. The model has 31,901 parameters to be trained.

Min-Max was used to normalise the data to a [0,1] range. Medium-term temporal patterns were learnt using a 60-day moving window to form input-output sequences. The data was divided 80/20 into the training (352 samples) and test (88 samples) set.

6.2 Training Process
The Adam optimiser was used to train the model with mean squared error (MSE) loss. A patience of 10 epochs was used to stop early in order to avoid overfitting. It reached a training loss of 0.0021 and validation loss of 0.0020 after training converged after 17 epochs, which reported good generalisation. The training trajectory is shown in Figure 8.

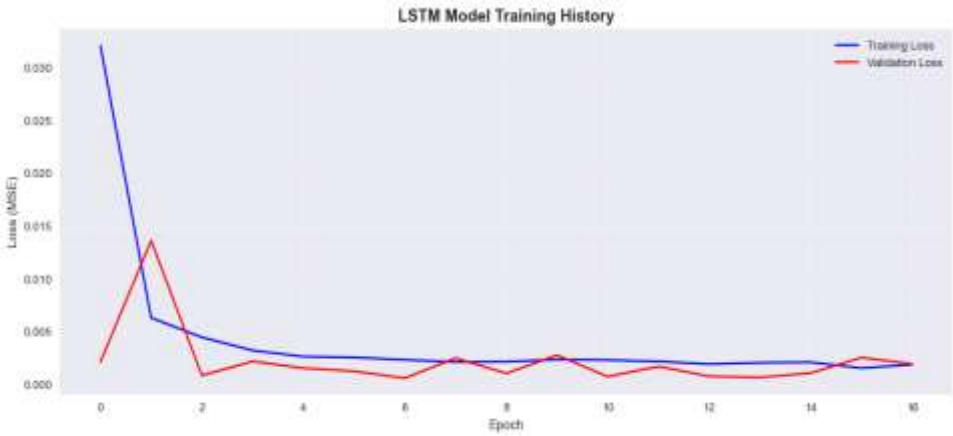

Figure 8: LSTM training and validation loss curves across epochs.

6.3 Test Set Evaluation

The LSTM model had an RMSE of 8.07, MAE of 6.36 and MAPE of 4.74 on the held-out test set. Although these measures are larger than the ARIMA in-sample values, this comparison is not quite fair: the ARIMA metrics represent in-sample (using training data), but the LSTM metrics are in-sample (using actual unseen test data). The LSTM estimates follow the real values of the test set fairly well, as provided in Figure 9, albeit with a small lag when the price changes fast-paced, which is an established fact of LSTM models.

Figure 9: LSTM predictions versus actual prices on the held-out test set.

6.4 24-Month Forecast

To determine the level of uncertainty, a Monte Carlo simulation strategy was implemented, and 50 iterative forecast runs were to be performed to provide long-range forecasts. The LSTM model gives a mean projection of about 159.63 in the 24 months future, which is set to stabilize at about 163.28 in February 2028. The forecast trend indicates that the prices are stabilizing, which is correlated with the fact that the LSTM has learned a mean-reverting or a plateauing trend of the later part of the training data. The confidence intervals are quite smaller in comparison to the ARIMA model and this could in part be attributed to the Monte Carlo methodology and not necessarily a reduction in uncertainty.

Figure 10: LSTM 24-month oil price forecast with 95% confidence intervals.

7. Comparative Analysis and Discussion

Figure 11 presents both forecasts on a single plot. The divergence between the two models is striking and instructive: the ARIMA model projects an aggressive continuation of the historical upward trend, while the LSTM model anticipates a plateau.

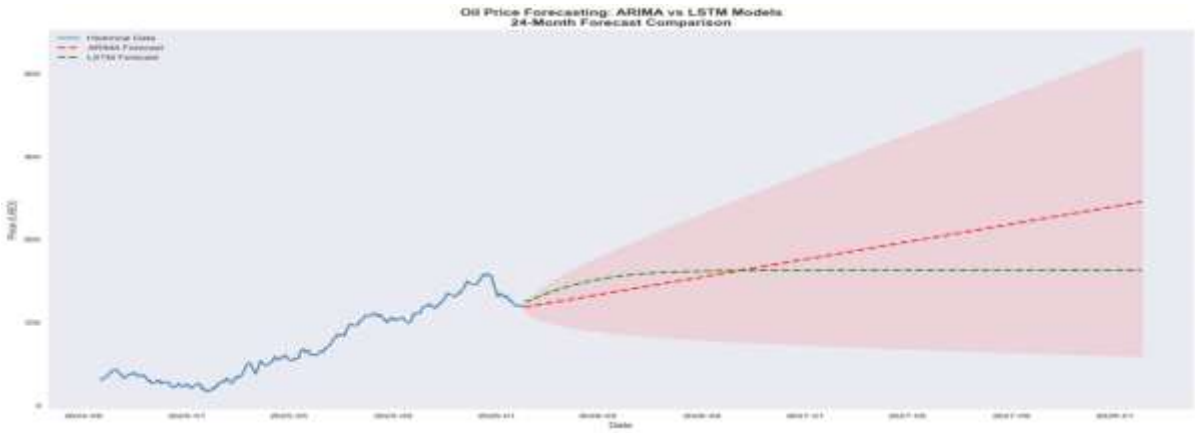

Figure 11: Combined 24-month forecast comparison — ARIMA versus LSTM with confidence intervals.

Metric	ARIMA(2,2,7)	LSTM (Stacked)
RMSE	1.8850	8.0742
MAE	0.8875	6.3631
MAPE	1.83%	4.74%
Evaluation Basis	In-sample	Out-of-sample (test)
Mean Forecast ($)	$182.06	$159.63
Final Forecast ($)	$245.63	$163.28
Model Type	Statistical / Linear	Deep Learning / Non-linear

Table 3: Comprehensive model performance comparison.

ARIMA model has higher in-sample fit measures which is not surprising since ARIMA parameters are directly optimised to reduce in-sample residues. The reason behind the steep upward forecast is that the d = 2 differencing parameter would be a continuation of the recent trend to an infinity. Although this can be suitable in case geopolitical or supply-side forces persist, it can produce unrealistically large forecasts in the case of a reverse or an equilibrium.

The LSTM model has been tested on unseen test data and this gives a better estimate of the predictive performance. Its forecast indicates that it will stabilise the prices at about $160, which is probably due to the deceleration in the growth of the prices that has been experienced later in the data. Deep learning models are also less conservative in their predictions because they learn from data as opposed to extending the history of patterns, unlike statistical models.

The other most important observation is the variation in confidence intervals. The intervals under the ARIMA model are greatly expanded in the long term, which reaches above 100 at the end of the forecast period showing how uncertain the long-term forecasts are. Conversely, the LSTM intervals are smaller, though, can have a reduced accuracy in capturing actual uncertainty due to the fact that Monte Carlo dropout may approximate model uncertainty, as well as fail to represent possible changes in the distribution entirely.

7.1 Supplementary Analysis: Seasonal Decomposition

To further explore the structure of the data an additive seasonal decomposition with a 30-day period (Figure 12) was conducted. Trend is a reflection of the rising trend whereas seasonal is an indication of slight cyclical movement of small magnitude compared to the total price level. The remaining part is random and this supports the sufficiency of the modelling strategies involved.

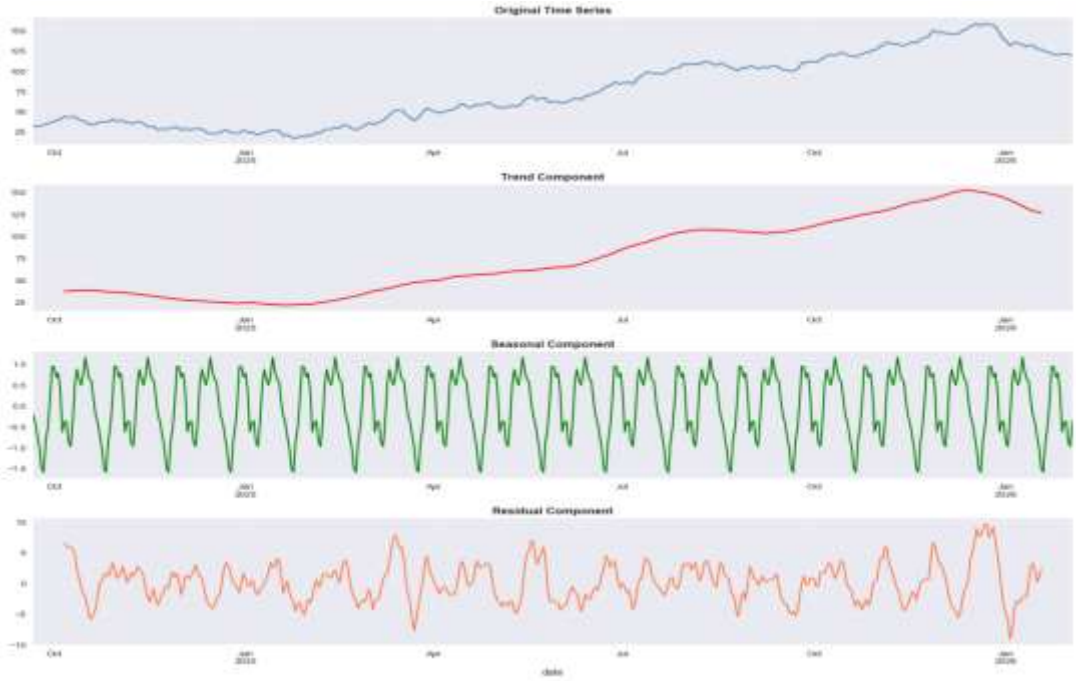

Figure 12: Additive seasonal decomposition of oil prices (period = 30 days).

8. Conclusions and Future Work

The present paper has introduced a comparison between the ARIMA and LSTM models in predicting the oil price based on the 2024- 2026 data. The ARIMA(2,2,7) model not only fitted reasonably well in-sample (MAPE = 1.83%) but also showed the uncertainty in a clear way by increasing confidence intervals. Even though the LSTM model revealed increased test error (MAPE = 4.74%), non-linear patterns were identified and potential price stabilisation was indicated.

No universal model is better than the other. ARIMA is better in short-term forecasting where trends are constant whereas LSTM may be better in longer-term forecasting where non-linear behaviour or change of structure is possible. Practically, the ensemble approach using the two models together may give more powerful predictions.

To enhance the models in the future, exogenous variables (e.g. OPEC production, geopolitical risk, macroeconomic indicators) can be added by the means of ARIMAX or multivariate LSTM models, hybrid architectures (including CNN-LSTM or Transformer), Bayesian uncertainty models, and rolling-window evaluation can be used to better evaluate out-of-sample performance.

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